

TeleVIP: On-site Person Removal and Context Distillation Platform for Dedicated Telepresence Experience

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ABSTRACT

For the application of telepresence robots, the on-site person usually has been regarded as a main element in remote interaction. However, in the context of a remote tour, the visual presence of an on-site person unexpectedly interrupting a robot's view may worsen the remote visitor's viewing experience. Moreover, the presence of a telepresence robot with its camera on may arise the privacy concerns of an on-site person. In this sense, we developed TeleVIP, a platform that provides telepresence applications with easy-to-use APIs to remove the figures of on-site people and selectively distill the useful information from the presence or context of on-site people, according to the application's demand. We implemented a working prototype of TeleVIP and performed pilot experiments in terms of both quantitative metrics and qualitative results, indicating the preliminary usability and feasibility of TeleVIP.

CCS CONCEPTS

• Human-centered computing → Mixed / augmented reality.

KEYWORDS

Telepresence Robot; Platform; Human Removal; Inpainting; Context Distillation

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1 INTRODUCTION

Telepresence robots have been widely researched in a variety of applications and contexts where interaction with a remote space occurs. Among those contexts, remote communications mediated by a telepresence robot have been researched for various types of social environments [6, 12, 14, 19]. In the context of remote communication, a person who locally exists in a remote space has been regarded as a main element for a remote participant to deal with.

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In other scenarios, however, the presence of an on-site person could act as a main hindrance rather than a main interactive element. A telepresence-mediated remote tour in a distant exhibit could be such a scenario. For the remote tour experience of visiting places like a distant art museum, the presence or movement of on-site people may unexpectedly or continuously interrupt the robot's view, annoying the telepresence user's viewing experiences. Although dealing with such on-site people would be common in real-world situations, there is an obvious social asymmetry between an on-site person and a remote visitor due to the limited mobility and expressiveness of the telepresence robot [2], making the remote visitor feel more pressured than an on-site person. Conversely, the on-site person may feel privacy concerns about the presence of the telepresence robot roaming around with its cameras on; it is little intuitively known to the on-site person about the identity or context of the remote user behind the telepresence robot.

In this sense, we develop TeleVIP, a telepresence platform that supports the features of selective removal or alterations of the on-site people on the remote user's perception. The design rationale of TeleVIP is that it could be deemed more beneficial to remove the figures of the on-site person from the view of the remote visitor, in favor of both parties' viewing experiences. TeleVIP accepts applications' requests which could be either simple removal of on-site people and/or an event subscription upon the on-site people's specific contexts (e.g., bodily gestures) to be met. Primarily, we apply the concept of diminishing perception on humans to the domain of telepresence by designing and implementing a remote tour system where on-site people are visually removed from the remote visitor's view. Furthermore, while removing the perceptual presence of on-site people, TeleVIP distills the useful information from the on-site people and renders it in alternative, less-annoying, and privacy-preserving forms. For example, a degree of localized crowdedness in an exhibition or even a high behavioral similarity therein may indicate a location of a potential attraction [9–11, 15, 17, 18]. Simply removing the whole presence of on-site people would result in such indications being lost.

For the remainder of this paper, we outline the conceptual and technical background. Then we introduce the overall architecture of TeleVIP and articulate the major components, followed by the report of the preliminary results from our pilot experiments, including basic performance metrics and qualitative examples.

2 RELATED WORK

Telepresence robots capable of networked video chat and self-mobility have been studied for their effectiveness in various social contexts such as hybrid work environments [19], classrooms [6],

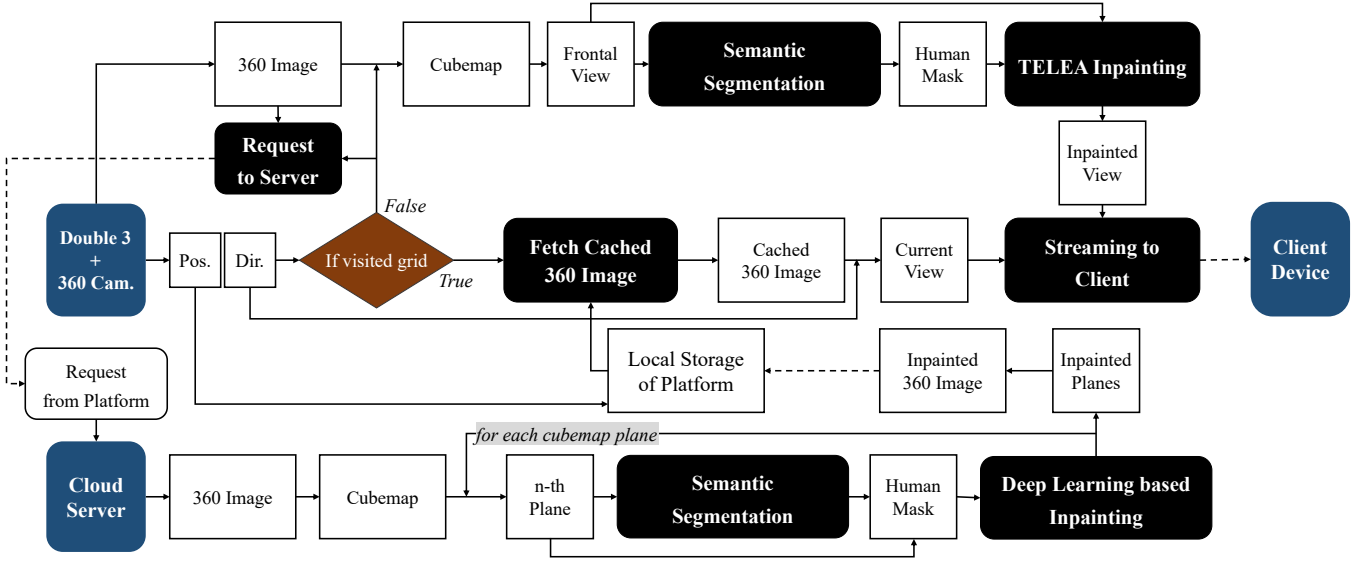


Figure 1: Architecture of TeleVIP

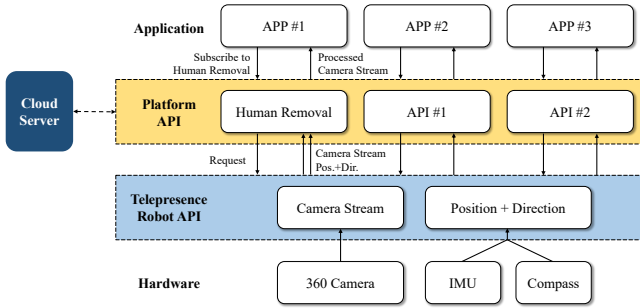


Figure 2: The Flow of Using APIs provided by the TeleVIP

and work-separated families [12]. Such telepresence robots are especially beneficial for remotely participating in a live social event since the user can freely roam around while visually interacting with the remote place in real-time [23].

The approach of purposefully removing or concealing the visual appearance of real-world scenes and objects has been researched in the domain of Diminished Reality (DR) [3]. Through DR, users perceive a visually simplified or even vacant version of the real environment. Several applications based on the concept of DR have been proposed. The target objects to be removed differ according to the purpose of each application. Humans have been also considered a target for the purpose of privacy protection for the on-site people [7], removing possible occlusions to the scene that the telepresence user tries to see [16], or better understanding the human-irrelevant scene properties [13]. Moreover, a set of technologies including real-time inpainting to make the implementation of such DR applications feasible have been researched [22].

3 DESIGN AND IMPLEMENTATION

3.1 TeleVIP

TeleVIP sits between the basic telepresence functions (e.g., locomotion, vision, sensing) and multiple telepresence applications, providing key APIs to ease the development and execution of telepresence applications demanding removal or alternative representation of on-site people’s presence and context. Figure 1 depicts the architecture of TeleVIP.

The key APIs of TeleVIP include human detection, human removal (with an option to inpaint), and event subscription APIs. Invoking one of these APIs registers a respective persistent service which keeps running until de-registered by the original caller application. For example, once the application invokes the human removal APIs with the inpainting option turned on, the camera stream from now on is directed to the real-time human segmentation unit followed by the real-time inpainting unit, where the output stream is delivered to the application.

The event subscription APIs allow the application to express various contextual conditions that it is interested in. Currently, TeleVIP supports the *people density* event and *bodily gesture* event, which detect a localized group of people above the application-specified spatial density [20, 21] and one of the pre-defined body poses and gestures (e.g., pointing with an arm) [24, 27], respectively.

We designed TeleVIP to operate in both on- and off-device modes so that it selectively offloads computationally heavy tasks to a high-end cloud server. Below, we explain the systematic operation of our human removal API and outline other APIs.

3.2 Human Removal API

In the human removal API, TeleVIP retrieves raw camera frames, and removes the figures of on-site people that appear on the frames. If a 360° camera is being used, TeleVIP refers to the user’s current viewport data to determine the rectangular portion to remove

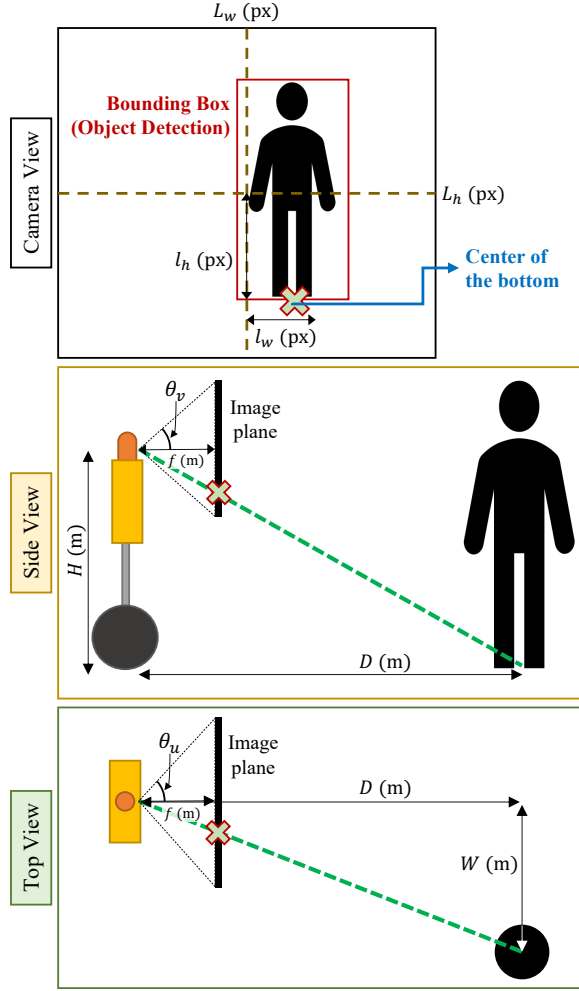


Figure 3: Calculation of On-site Person's Location based on Object Detection

on-site person appearances therein. Two main techniques are employed to remove on-site person appearances from the camera view: semantic segmentation and image inpainting. Human masks are extracted from the semantic segmentation, and the masked regions are filled through the image inpainting technique. To regulate the overall computational workload for these processings, TeleVIP caches each inpainted frame along with the robot's self-odometry information on its local storage, for possible reuse upon the robot re-visiting this scene within a time limit of cache-staleness threshold. TeleVIP divides the space into equally sized square grids for cache management.

Considering the trade-offs between latency and the quality of the inpainted results, TeleVIP employs multiple segmentation and inpainting libraries each specializing in on- or off-robot execution. Upon the robot entering a new region that has never been cached or was cached too past beyond the staleness limit, low-latency models are preferred for immediate delivery to the remote user. For this purpose, TeleVIP has locally executable light-weight semantic

segmentation (Tensorflow BodyPix library) and inpainting (TELEA algorithm [25]), respectively. Otherwise, once-visited scenes are referred to the off-device cloud server to refine the inpainting quality by applying a heavier image inpainting model [26].

3.3 Crowdedness Detection API

The presence or certain context of on-site people may convey useful information to the telepresence application even though those people's visual appearances are removed. Currently, TeleVIP supports distilling the degree of localized crowdedness and certain predefined bodily gestures from the on-site people.

Below, we detail the crowdedness detection as an example. To determine localized crowdedness, TeleVIP needs to detect a group of people and their spatial density, which requires the distance information of each person detected in the scene. Upon using a camera with a depth sensor available, this could be trivially done. Upon using a 360° camera which does not support all-around depth information, the spatial location of each person can be still estimated under two assumptions: the robot is exploring on a flat ground, and the object detection model is accurate enough that the bottom side of the bounding box estimated is well aligned with a foot of the corresponding person.

Let's assume that a person is located at an arbitrary location $(D(m), W(m))$ with respect to the 360° camera of a focal length $f(m)$, where the camera is installed at the telepresence robot's height $H(m)$ (Figure 3). Through this formulation, we can develop two proportional expressions:

$$D : H = D - f : H - l_h \times \frac{2f \tan \theta_v}{L_h}$$

$$D : W = f : l_w \times \frac{2f \tan \theta_u}{L_w}$$

Then, we can derive the location of the person, i.e., D and W , like below:

$$D = \frac{HL_h}{2l_h \tan \theta_v}$$

$$W = \frac{HL_h l_w \tan \theta_u}{L_w l_h \tan \theta_v}$$

Once the location of every on-site person has been calculated, we determine the spatial density of a localized group of people. If a group density remains above the application-specified value for more than 3 seconds (an empirically chosen time window), TeleVIP notifies the application that a crowdedness event has happened, along with the on-site location information. The application may indicate this event to the telepresence user's interface, e.g., rendering a graphical icon associated with the detected on-site location.

4 RESULT

Based on the design described in Section 3, we implemented a working prototype of TeleVIP, and performed pilot experiments to evaluate the quantitative metric and qualitative results. The prototype was built in a local environment on Double 3 [1], which is a telepresence robot commercially being on sale.

We first evaluated the results of human removal in terms of latency metric. In Double 3, the latency on semantic segmentation and image inpainting has been quantitatively evaluated. We run

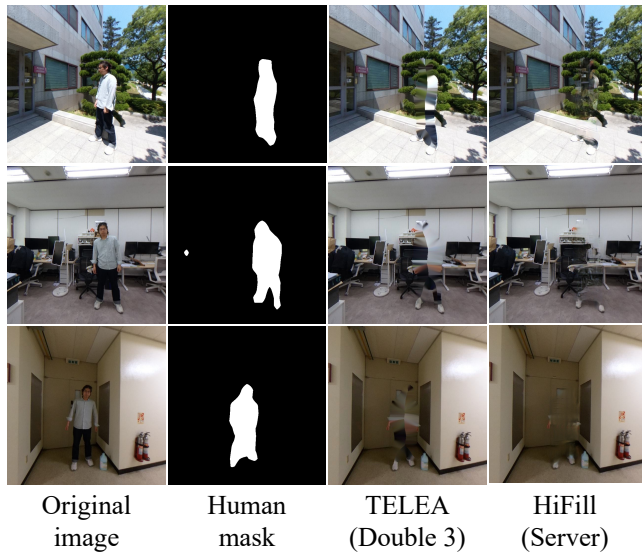


Figure 4: Qualitative Comparison on Inpainting Results between the Local Environment of Double 3 and the Cloud Server

MobileNet [8] provided by BodyPix library for semantic segmentation, and run TELEA algorithm [25] for image inpainting. In actual implementation, the result of the semantic segmentation model has been dilated by 3×3 kernel to deal with inaccurate parts of a mask, which has been also included in a latency metric for semantic segmentation. We run the models on 6 images, frontal views captured by Ricoh Theta V. As a result, each semantic segmentation and image inpainting showed an average latency of 35.3 ms and 39.1 ms respectively, which indicates that the responsiveness of human removal operations can be guaranteed.

We also evaluated the latency of inpainting method run on the high-end cloud server. We use HiFill [26] model for the server-side image inpainting, and run inference on a single GPU, NVIDIA RTX 6000 Ada. Although we didn't use HiFill [26] model on Double 3, we also checked the latency of running the same model on an embedded GPU of Double 3, Jetson TX2, to confirm whether it is rational to off-load the work on the server. As a result, the server showed a much shorter latency (91.0 ms) than the local environment of Double 3 (837 ms), which indicates that offloading the work to the high-end server is a rational choice.

Lastly, we qualitatively compared the results of human removal as illustrated in Figure 4, and the server-side results showed a significantly better inpainting quality than the results from Double 3 as we expected in the design of TeleVIP.

5 DISCUSSION AND CONCLUSION

In this paper, we developed a telepresence platform named TeleVIP which selectively provides a processed camera stream where all figures of on-site people are removed and useful information is distilled from a camera stream to applications according to their requests. In addition, we implemented a working prototype of TeleVIP on the local environment of Double 3 with APIs of human removal

and a degree of localized crowdedness, and presented quantitative and qualitative results on the processing of TeleVIP. We expect that TeleVIP can be functionally more generalized for remote tour applications by adding more distillable features in a set of APIs, and further be a cornerstone for a remote tour system to apply the concept of diminishing perceptions on humans. To effectively support a semi-interactive scenario, TeleVIP would be extended with additional APIs to assess the interaction probability for each person (through a rule-based, history-based, or a proactive probing [4]), so that the human removal API could be invoked selectively by the application or direct user input [5].

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